Xiangyu Ding

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Research interests

Integer partitions; q-series; Bailey chains and lattices; truncated theta series; Rogers–Ramanujan-type identities.

Education

Integrated graduate-to-PhD programme in Applied Mathematics, Nankai University | September 2021 – June 2027 (expected)

Doctoral stage since September 2023

Supervisor: Professor Lisa Hui Sun

Papers

[1] X. Ding and L. H. Sun, Truncated theta series from the Bailey lattice. *Advances in Applied Mathematics* 167 (2025), 102884. arXiv:2403.11608.

[2] X. Ding and L. H. Sun, On Merca's stronger conjecture for the truncated Jacobi triple product series. submitted. arXiv:2411.13818.

[3] X. Ding and L. H. Sun, A bijective partition-theoretic interpretation for the Andrews-Merca truncated pentagonal number theorem. submitted.

[4] X. Ding and L. H. Sun, A combinatorial proof for the positivity of the normalized Jacobi triple product tails. *Journal of Combinatorial Theory, Series A*, accepted. arXiv:2606.27507.

[5] X. Ding and L. H. Sun, A symmetrized B_2 Bailey lemma and Rogers–Ramanujan-type identities. preprint.

[6] I. D. Chen, X. Ding, Q. Guo and L. H. Sun, The extension of Ballantine and Merca's conjecture on 6-regular partitions. submitted.

Awards

Shiing-Shen Chern Academic Newcomer Scholarship, Nankai University, 2025

陈省身学术新人奖学金

Academic activities

Participant, 36th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2024), Ruhr-Universität Bochum, Germany, 22–26 July 2024.

Research methods

Bailey methods and basic hypergeometric transformations, including conjugate WP–Bailey pairs and finite kernel constructions.

Explicit partition bijections using Durfee rectangles, staircase operations and Gaussian refactorization; sign-reversing involutions, minimal excludants and Frobenius-symbol injections.

Exact enumeration of restricted partitions, residue-class coefficient analysis, and effective circle-method estimates for nonmodular infinite products.

Postdoctoral research interests

New Bailey roots and lattice directions; combinatorial interpretations of two-family multisums; connections between partition identities and specialized character formulas.

References

Professor Lisa Hui Sun (PhD supervisor)
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